

# A three-way clustering method based on an improved DBSCAN algorithm

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## HIGHLIGHTS

- A three-way clustering method 3W-DBSCAN is proposed.
- The representation of clustering results is consistent with human cognitive thinking.
- Experiments show that 3W-DBSCAN has a good performance in clustering.

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## ABSTRACT

Clustering is a fundamental research field and plays an important role in data analysis. To better address the relationship between an element and a cluster, a Three-Way clustering method based on an Improved DBSCAN (3W-DBSCAN) algorithm is proposed in this paper. 3W-DBSCAN represents a cluster by a pair of nested sets called lower bound and upper bound respectively. The two bounds classify objects into three status: belong-to, not belong-to and ambiguity. Objects in lower bound certainly belong to the cluster. Objects in upper bound while not in the lower bound are ambiguous because they are in a boundary region and might belong to one or more clusters. Objects beyond the upper bound certainly do not belong to the same cluster. This clustering representation can well explain the clustering result and consist with human cognitive thinking. By improving similarity calculation, improved DBSCAN is presented to obtain initial clustering results, then three-way decision strategies are used to acquire the positive and boundary regions of a cluster. Three benchmarks Accuracy ( $Acc$ ), F-measure ( $F_1$ ),  $NMI$  and ten datasets including three synthetic datasets, three UCI datasets and four shape datasets are used in experiments to evaluate the effectiveness of 3W-DBSCAN. Experimental results suggest that 3W-DBSCAN has a good performance and is effective in clustering.

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## 1. Introduction

As a powerful data analysis tool, clustering plays an important role in data mining, which is widely used in various areas such as image analysis [1,2], network structure analysis [3–6], bioinformatics [7–9] and so on. The objective of clustering is to group similar objects into the same cluster while dissimilar objects into different clusters [10–12]. It is an unsupervised learning method without any prior information to find potential similar patterns from a dataset [13–16].

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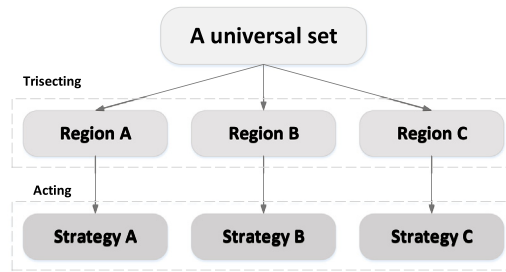


Fig. 1. Trisecting-and-acting model [19,20].

Many existing clustering methods assume that each object must be assigned to exactly one cluster, which results in the type of hard clustering, namely an object only belongs to one cluster. However, in many applications, clusters may not necessarily have clear and sharp boundaries [17], i.e. an object can be assigned to any cluster or two or more clusters in the same time. It is difficult to give a clear hard clustering result under the condition of incomplete or inaccuracy information. For example, in an interest network, a member may have multiple interests and it may belong to different clusters. If we deal with these datasets on hard clustering, it will lead to a higher error rate or decision risk. It shows that hard clustering methods may not sufficiently explain the relationship between an element and a cluster.

Whether an element belongs to a cluster or not is determined by its position in the dataset. It might fully belong to one cluster, fully not belong to one cluster, belong to several clusters at the same time or as a noisy point. From this perspective, we may use three types to describe the relationship between an element and a cluster: belong-to, not belong-to, and ambiguous. In order to represent the three types of relationships, we adopt a notion of three-way clustering [18], as an alternative to conventional two-way clustering. Three-way clustering divides the whole discussing region into three parts: positive region (POS), negative region (NEG) and boundary region (BND). Different regions adopt different ways to manage respectively. For one cluster, if an object is in a positive region, we can make a decision of acceptance, namely it belongs to the cluster. If an object is in a negative region (NEG), we can make a decision of rejection, i.e. it does not belong to the cluster. If an object is in a boundary region, we do not make a decision immediately, namely non-commitment or deferment, until we have enough information about these problems to make a right judgement.

The idea of three-way clustering originated from three-way decision [19,20]. Three-way decision is an extension of the commonly used binary-decision models. Three-way decision theory believes that people usually make decisions based on current available information and evidence, however, if the information is insufficient or weak, it might be impossible to make either a positive or a negative decision. Thus one may choose an alternative way to make a decision, namely neither yes or no, which is also called a deferment decision and requires further judgement. Therefore, the basic idea of three-way decision is to divide a universal set into three pair-wise disjoint regions and to make three types of decisions accordingly, as shown in Fig. 1. This tri-partition methodology provides flexible ways for complex problem solving and information processing [21,22]. The idea of three-way decision has been introduced into a variety of theoretical methods, like evidence theory [23,24] and fuzzy set [25–27]. The wide sense of three-way decision offers a new understanding and hence many new research branches have appeared like three-way decision support [28,29]. Three-way decision theory has also been widely used in many applications of uncertain information [30–32].

Inspired by the idea from three-way decision, Yu [33] applied it to clustering and proposed the concept of three-way clustering. The concept of three-way clustering was also considered by Lingras [34,35] and Yao et al. [36,37]. Recently Afridi et al. [38] constructed a three-way clustering approach for handling missing data. Wang et al. [17] put forward a general framework of three-way clustering (CE3) in view of erosion and dilation from mathematical morphology. Yu et al. [18,39] also developed new clustering approaches for more uncertain situations such as dynamic, incomplete data or multi-source data.

DBSCAN [40], or density-based spatial clustering of applications with noise, is a typical successful density-based clustering method. It can not only find arbitrary shape clusters and handle noise points, but also can detect the number of clusters naturally. However, a crucial problem with DBSCAN is that it faces the challenge of finding clusters with different densities using a global density threshold [41–43]. To overcome this, many variants of DBSCAN and methods have been proposed. For example, OPTICS [44] sorts the neighborhood points according to the density, and then draws a “reachability” plot to find clusters of different densities. SNN based clustering [45] uses the number of shared nearest neighbors between two points in replace of distance measure in DBSCAN, thus can ignore objects’ true density distribution. DP [46,47] can determine cluster centers and group objects based on local density and relative distance.

In this paper, we apply three-way decision theory and put forward a Three-Way clustering method based on an Improved DBSCAN algorithm, named 3W-DBSCAN for short. Not like a single set to express a cluster in two-way clustering, we represent a three-way cluster by a pair of nested sets. We also use a typical representative density-based clustering method DBSCAN [40] and improve its similarity calculation to cluster the dataset. It can find the number of clusters based on the estimated density distribution and has several advantages. Firstly, it can discover clusters of different sizes and

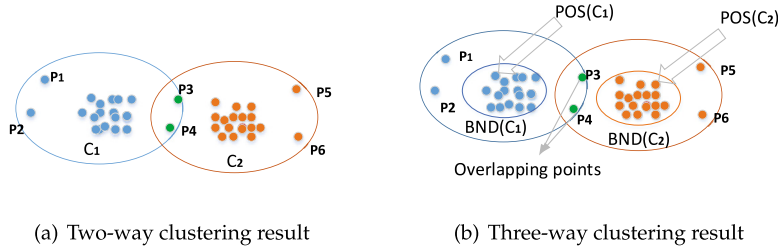


Fig. 2. Representation of two-way clustering and three-way clustering.

shapes. Secondly, it need not know the number of clusters in advance. Thirdly, with a global density threshold, it can identify clusters with varied densities.

The organization of the rest paper is as follows. Section 2 presents the proposed algorithm 3W-DBSCAN including the clustering representation by three-way decision, improved DBSCAN and three-way clustering processing. In Section 3, we empirically evaluate the performance of 3W-DBSCAN. Section 4 is the conclusion of this paper.

## 2. The 3W-DBSCAN method

### 2.1. Clustering representation by three-way decision

Suppose  $C = \{C_1, C_2, \dots, C_k\}$  is a family of clusters, and  $V = \{x_1, x_2, \dots, x_n\}$  is a finite non-empty universe of objects.  $x_i$  is an object which has  $h$  attributes,  $x_i = (x_{i1}, x_{i2}, \dots, x_{ih})$ , where  $x_{ij}$  denotes the value of  $j$ th attribute of object  $x_i$ ,  $i = \{1, 2, \dots, n\}$  and  $j = \{1, 2, \dots, h\}$ .

**Definition 1.** A cluster is depicted by a pair of nested sets [33]:

$$C_i = [\underline{C}_i, \overline{C}_i] \quad (1)$$

where  $\underline{C}_i$  is the lower bound of the cluster and  $\overline{C}_i$  is the upper bound of the cluster,  $\underline{C}_i \subseteq \overline{C}_i \subseteq V$ . The set  $\underline{C}_i$ ,  $\overline{C}_i - \underline{C}_i$  and  $V - \overline{C}_i$  can form into three regions of cluster  $C_i$  as positive, boundary and negative regions, respectively. The three-way clustering representation is given as:

$$\begin{aligned} POS(C_i) &= \underline{C}_i \\ BND(C_i) &= \overline{C}_i - \underline{C}_i \\ NEG(C_i) &= V - \overline{C}_i \end{aligned} \quad (2)$$

Objects in  $POS(C_i)$  definitely belong to the cluster  $C_i$ , objects in  $NEG(C_i)$  definitely do not belong to the cluster  $C_i$ . Objects in  $BND(C_i)$  imply that their relationship with clusters is ambiguous, we need more information to make judgements and thus they may belong to one or more clusters.

In this paper, the subsets of cluster satisfy the following conditions.

- (a)  $POS(C_i) \neq \emptyset$
- (b)  $POS(C_i) \cap BND(C_i) = \emptyset$
- (c)  $BND(C_i) \cap NEG(C_i) = \emptyset$
- (d)  $POS(C_i) \cap POS(C_j) = \emptyset, i \neq j$
- (e)  $POS(C_i) \cup BND(C_i) \cup NEG(C_i) = V$

Under the three-way representation of a cluster, the family of clusters can be depicted by interval sets as:

$$C = \{[\underline{C}_1, \overline{C}_1], [\underline{C}_2, \overline{C}_2], \dots, [\underline{C}_k, \overline{C}_k]\} \quad (3)$$

Following is an example to simply demonstrate the difference between two-way clustering and three-way clustering. Given a dataset including two concentrated areas and six relatively discrete points, if a traditional two-way clustering algorithm is used, as shown in Fig. 2(a), six discrete points  $p_1, p_2, \dots, p_6$  will be strictly assigned to each cluster. However, focus on  $p_3$  and  $p_4$ , intuitively, their distance to  $C_1$  and  $C_2$  are almost the same and it seems unreasonable to assign them to either one of the two clusters, it may be more appropriate to regard them as overlapping areas between  $C_1$  and  $C_2$ . Besides,  $p_1$  and  $p_2$  are relatively close to  $C_1$ , but they are far from the core of  $C_1$ , namely their relationship with  $C_1$  is not as strong as the points in core of  $C_1$ , the same situation also happens on  $p_5$  and  $p_6$ . If we apply three-way decision method on the same dataset and can obtain the three-way clustering result in Fig. 2(b), in which six discrete points are assigned

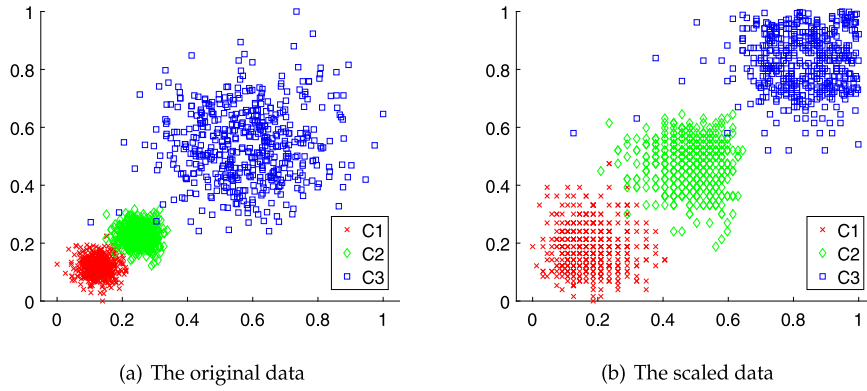


Fig. 3. Original and scaling data distribution.

to the boundary region  $BND$  of each cluster. It means that their relationship with clusters is loose, not as strong as the points in positive regions  $POS(C_1)$  and  $POS(C_2)$ . They may belong to one cluster or may be in overlapping areas between different clusters. This representation is more in line with people's cognition than two-way clustering.

## 2.2. DBSCAN clustering algorithm

As one of the successful density-based clustering algorithm, DBSCAN can find several clusters based on the estimated density distribution. It need not know the cluster number in advance and can find shaped clusters. The basic idea of DBSCAN is illustrated as follows. Starting from a random unvisited point  $p$ , DBSCAN retrieves all points in the  $\epsilon$ -neighborhood of  $p$ , where  $\epsilon$  is the maximum radius of the neighborhood from  $p$ . The density threshold  $MinPts$  is the minimum number of points required to form a dense region. Point  $p$  is a core point if at least  $MinPts$  points are within distance  $\epsilon$ . If  $p$  is a core point, all points in its  $\epsilon$ -neighborhood are classified into the same cluster as  $p$ . For each point in the cluster, DBSCAN finds all density-reachable points and adds them into the same cluster. If a point  $q$  is density-reachable from other core points, but its  $\epsilon$ -neighborhood is less than  $MinPts$ , then it is the border point belonging to the cluster as well. If a point is not reachable from any other point, it is a noisy point or outlier. DBSCAN achieves the clustering process by extracting clusters sequentially. Repeating this process until no new density-reachable points are found, a final cluster is obtained. DBSCAN divides a set of points into three types: core point with high density, border point with low density and noise. The definition of three types of points is shown as follows.

**Definition 2** (Three Types of Points). Given any two points  $x$  and  $y$ ,  $d(x, y)$  is the similarity between them,  $\Gamma_\epsilon(x)$  is the  $\epsilon$ -neighborhood of  $x$ , where  $\Gamma_\epsilon(x) = \{y \in V | d(x, y) \leq \epsilon\}$ .  $\rho(x) = |\Gamma_\epsilon(x)|$  is the density value of  $x$ . Three types of points are depicted by the type function  $S(x)$ , it is defined as:

$$S(x) = \begin{cases} 1 & \text{core point with } \rho(x) \geq MinPts \\ 0 & \text{border point with } 1 < \rho(x) < MinPts \\ -1 & \text{noise with } \rho(x) = 1 \end{cases} \quad (4)$$

## 2.3. Improved similarity calculation in DBSCAN

As mentioned above, DBSCAN has several advantages, however, it cannot find clusters with varied densities. Many efforts have been done to improve it [41,48]. Recently, Zhu et al. [49] provided a multi-dimensional distance scaling method (DScale), which can be regarded as a data pre-processing technique. Its basic idea is scaling the computed distance between every two points, which causes the estimated density of each scaled point to change. It also has been proved that in case of a single threshold, DScale allows an existing density-based cluster algorithm to find all clusters with varied densities [49].

Inspired by this idea, we improve the similarity calculation in traditional DBSCAN, so that it can identify clusters of different densities. The improved DBSCAN is defined as follows.

**Definition 3** (The Scaling Function). For a dataset  $V = \{x_1, x_2, \dots, x_n\}$  of  $n$  objects, each object has  $h$  features,  $x_i \in R^h$ . Given a computed similarity matrix  $D = [d(x, y)]$ ,  $d_{max} = \max_{x, y \in V} d(x, y)$ . The scaling function is defined as:

$$r(x) = \left( \frac{|\Gamma_\eta(x, d)|}{n} \right)^{\frac{1}{h}} \times \frac{d_{max}}{\eta} \quad (5)$$

where the similarity matrix  $D$  can be obtained by Euclidean distance,  $\Gamma_\eta(x, d)$  is the  $\eta$ -neighborhood of  $x$ ,  $\Gamma_\eta(x, d) = \{y \in V | d(x, y) \leq \eta\}$ ,  $\eta \in (0, 1)$  is a parameter of the size of neighborhood radius. The scaling function of Eq. (5) is to make the data approximately uniformly distributed in the scaled  $\eta$ -neighborhood.

**Definition 4** (*The Scaled Distance*). Given a computed distance  $d(x, y)$ , the scaled distance  $d'(x, y)$  is obtained by the scaling function  $r(x)$  and defined as:

$$d'(x, y) = \begin{cases} d(x, y) \times r(x) & \text{if } y \in \Gamma_\eta(x, d) \\ (d(x, y) - \eta) \times \frac{d_{\max} - \eta \times r(x)}{d_{\max} - \eta} + \eta \times r(x) & \text{if } y \in V \setminus \Gamma_\eta(x, d) \end{cases} \quad (6)$$

The scaled distance is computed in two cases, when  $y \in \Gamma_\eta(x, d)$ , Eq. (6) is a linear scaling, hence the number of points in  $\Gamma_\eta(x, d)$  remains the same as in  $\Gamma_{\eta \times r(x)}(x, d')$ , otherwise when  $y \in V \setminus \Gamma_\eta(x, d)$ , Eq. (6) is actually a min-max normalization to keep the same sample rank.

The mechanism of scaling is the same to density-ratio based clustering [41]. We take a simple example to illustrate it, Fig. 3(a) is a mixture of three Gaussian distributions from Ref. [41], where  $C_1$  and  $C_2$  are relatively with high density and  $C_3$  is with low density. Fig. 3(b) shows the new distribution on scaled data of (a), we can find that the density gaps between three clusters are reduced and hence DBSCAN can detect three clusters using a single threshold. In summary, the function of Eqs. (5)–(6) is to make the data approximately uniform according to the bandwidth ( $\eta$ ), after transformation, the densest areas in the original data are expanded in the scaled data, while the sparsest areas are shrunk. As a result, different cluster modes are more distanced from each other, and the gap between different clusters' boundaries are enlarged in the scaled data, therefore based on the scaled data, DBSCAN allows for a single threshold to find clusters with varied densities.

From above definition, we can obtain the scaled similarity matrix  $D' = [d'(x, y)]$  by Eq. (6). DBSCAN is then implemented on the basis of  $D'$  to find clusters. Its result is represented as  $C = \{C_1, \dots, C_i, \dots, C_k\} \cup No(C)$ ,  $k$  is the number of clusters identified by DBSCAN,  $No(C)$  includes all noisy objects from DBSCAN clustering result.

#### 2.4. Three-way clustering processing

Not like a single set used in two-way clustering, 3W-DBSCAN applies a pair of sets  $C_i = [C_i, \bar{C}_i]$  to depict a cluster. Therefore, after obtaining clustering result by Improved DBSCAN, three-way clustering processing is then implemented, which mainly consists of three strategies: (1) Determine the positive and boundary region of  $C_i$ , (2) Expand the boundary region of  $C_i$  and determine them in the overlapping regions whether or not and (3) Get more information to assign noisy points to boundary region of clusters.

Strategy 1: Determine the positive and boundary region of  $C_i$

For each cluster  $C_i$  acquired by improved DBSCAN, in order to construct the positive region  $POS(C_i)$  and the boundary region  $BND(C_i)$ , the basic idea is to compare objects' density with the threshold  $MinPts$ . If an object  $x$  belongs to a high density field of  $C_i$ , i.e.  $S(x) = 1$  in Eq. (4), thus it is a typical element of  $C_i$  and naturally should be assigned to the positive region. If an object  $x$  is in a low density field, i.e.  $S(x) = 0$  in Eq. (4), it is possible that it may belong to one or more clusters,  $x$  will be assigned to the boundary region. Therefore, the positive and boundary region of  $C_i$  can be constructed as follows:

$$\begin{aligned} POS(C_i) &= \{x | S(x) = 1, x \in C_i\} \\ BND(C_i) &= \{x | S(x) = 0, x \in C_i\} \end{aligned} \quad (7)$$

Actually, for a cluster  $C_i$ ,  $POS(C_i)$  is the set of all core points and  $BND(C_i)$  is the set of all border points.

Strategy 2: Process overlapping objects to expand boundary regions

In Strategy 1, border objects are assigned to single corresponding boundary region, but we need to further deal with them because they may be the members of other clusters, i.e., overlapping objects. The same situation may also happen at noisy points. The following strategy is adopted: for one point  $x$  satisfying  $S(x) = 0$  or  $S(x) = -1$ , if a neighbor of  $x$  in  $\epsilon$ -neighborhood size belongs to one other cluster  $C_i$ ,  $x$  is considered as an overlapping object and then add  $x$  to the set  $BND(C_i)$ . Hence the boundary region  $BND(C_i)$  is updated as:

$$BND(C_i) = BND(C_i) \cup \{(x | C_i \cap \Gamma_\epsilon(x) \neq \emptyset)\} \quad (8)$$

Strategy 3: Get more information to assign remaining noisy points to boundary region of clusters.

For the remaining noisy points unassigned in Strategy 2, we want to further process and assign them to the proper clusters. Therefore, for each noisy point  $x$ , we adopt the following strategy to tackle with it. Set  $AllPOS$  denote the set of all core objects, i.e.,  $AllPOS = \bigcup_{i=1}^k POS(C_i)$ . The nearest core neighbor of  $x$ , represented as  $NCN(x)$ , is measured by satisfying the minimum distance from  $x$  to another point in the set  $AllPOS$ , as shown in Eq. (9).

$$NCN(x) = y \leftarrow \arg \min_{y \in AllPOS} d(x, y), S(x) = -1 \quad (9)$$

Then the noise  $x$  is assigned to the boundary region of the same cluster as its nearest core neighbor  $NCN(x)$ :

$$\text{if } NCN(x) \in POS(C_i), \text{ then } BND(C_i) = BND(C_i) \cup x \quad (10)$$

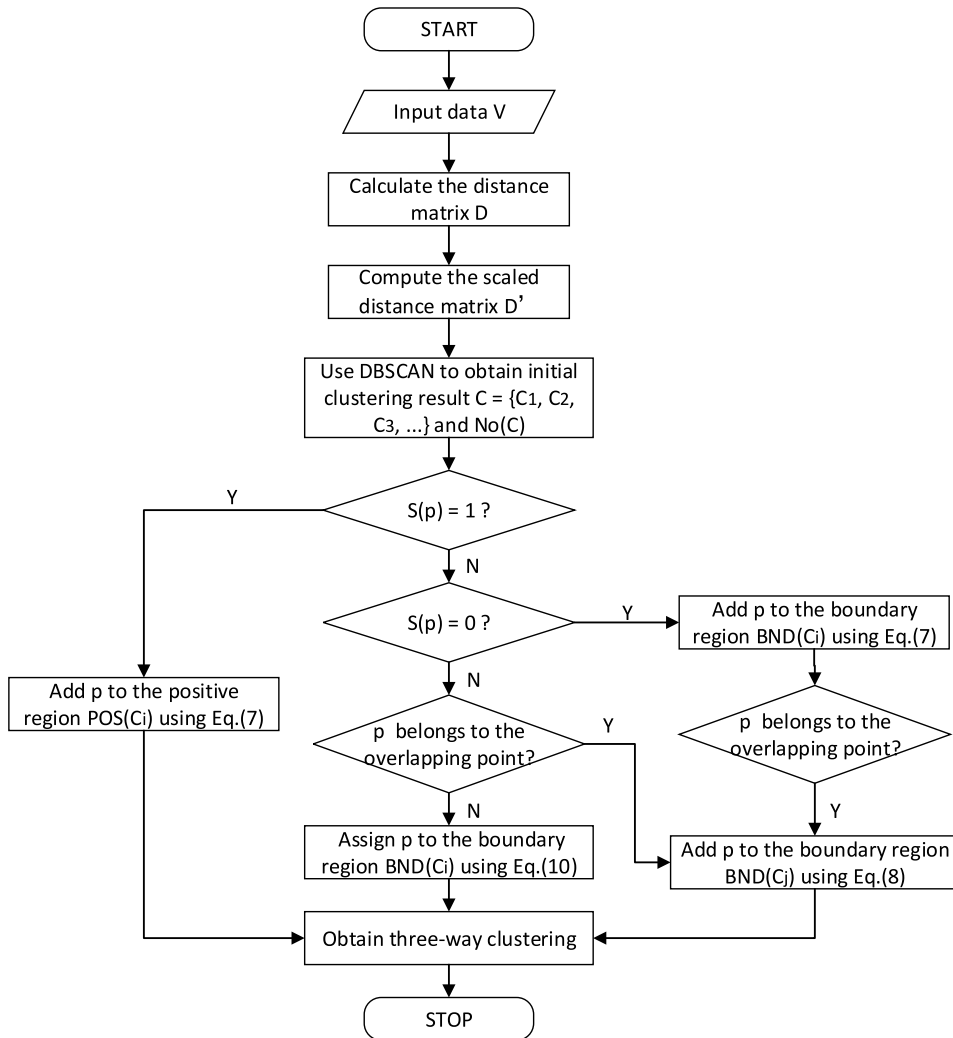


Fig. 4. The flowchart of 3W-DBSCAN.

Table 1

Data properties of 10 datasets.

| Type              | Dataset     | Data size | Dimension | Clusters | Source |
|-------------------|-------------|-----------|-----------|----------|--------|
| Synthetic dataset | 3L          | 560       | 2         | 3        | [49]   |
|                   | 4C          | 1250      | 2         | 4        | [49]   |
|                   | S1          | 5000      | 2         | 15       | [50]   |
| UCI dataset       | IRIS        | 150       | 4         | 3        | [51]   |
|                   | Glass       | 214       | 9         | 6        | [51]   |
|                   | Seeds       | 210       | 7         | 3        | [51]   |
| Shape dataset     | Pathbased   | 300       | 2         | 3        | [48]   |
|                   | Aggregation | 788       | 2         | 7        | [52]   |
|                   | Flame       | 240       | 2         | 2        | [52]   |
|                   | Compound    | 399       | 2         | 6        | [48]   |

Fig. 4 shows the flowchart of 3W-DBSCAN. The whole procedure of 3W-DBSCAN is summarized in Algorithm 1 and the detailed time complexity is analyzed: Line 1 is to calculate the distance matrix which needs  $O(n^2)$ , in Line 2, since the algorithm needs to manage with each point, the basic complexity is  $O(n^2)$ . The DBSCAN algorithm in Line 3 also

needs  $O(n^2)$  in the worst case. Suppose  $n_1$  is the number of clustered samples and  $n_2$  is the number of noise after running DBSCAN, i.e.,  $n_1 + n_2 = n$ . Line 4 needs to scan the value of type function for  $n_1$  points, it costs  $O(n_1)$ , the complexity of Line 5 is  $O(n)$  at most. In Line 6, to find the nearest core neighbor, the algorithm needs  $O(n_1 n_2)$  mostly. Generally, improved DBSCAN needs  $O(n^2)$  and three-way clustering processing costs  $O(n_1 n_2)$ ,  $O(n_1 n_2) < O(n^2)$ , hence the time complexity of 3W-DBSCAN is  $O(n^2)$ .

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**Algorithm 1:** The proposed algorithm 3W-DBSCAN
 

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**Input :** A dataset  $V = \{x_1, x_2, \dots, x_n\}$

**Output:**  $C = \{\underline{C}_1, \overline{C}_1\}, \{\underline{C}_2, \overline{C}_2\}, \dots, \{\underline{C}_k, \overline{C}_k\}$

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- 1 Calculate the distance matrix  $D$ ;
  - 2 Obtain the scaled distance matrix  $D'$  using Eqs. (5)–(6);
  - 3 Based on  $D'$ , use DBSCAN algorithm to cluster as  $C = \{C_1, C_2, \dots\} \cup No(C)$  ;
  - 4 Determine the positive and boundary region using Eq. (7);
  - 5 Process overlapping objects to expand boundary regions using Eq. (8);
  - 6 Assign noises to boundary region of clusters using Eqs. (9)–(10);
- 

### 3. Experimental evaluation

#### 3.1. Datasets

In this section, we present experiments to evaluate the effectiveness of 3W-DBSCAN using 3 synthetic datasets, 3 UCI datasets and 4 shape datasets. Table 1 shows the properties of 10 datasets. Note that among synthetic datasets, 3L is a 2-dimensional data containing three elongated clusters and 4C contains four clusters including three Gaussian clusters and one elongated cluster. S1 is with 5000 instances and 15 Gaussian clusters. IRIS, Glass and Seeds are all with multidimensional attributes. Pathbased is a shaped dataset where two Gaussian and circular clusters are within connections, Flame is also featured with the connected distribution. Aggregation contains seven circular clusters with similar densities. Compound creates a challenge for clustering algorithms as it is formed by different shapes and densities [48].

#### 3.2. Evaluation measures

We use three traditional indices including Accuracy (Acc) [17], F-measure ( $F_1$ ) [49] and NMI [53] to evaluate the performance of 3W-DBSCAN. The brief introduction of three indices is given below.

##### 1. Accuracy (Acc)

$$Acc = \sum_{i=1}^k \frac{n_i}{n} \quad (11)$$

where  $n_i$  is the number of correct objects which are predicted by clustering algorithm comparing with true clusters.  $n$  is the total number of objects. A higher  $Acc$  indicates a better clustering result.

##### 2. F – measure( $F_1$ )

$F_1$  [49] is computed by the harmonic mean of precision score  $P$  and recall score  $R$ , and the overall  $F_1$  is the unweighted average over all clusters.

$$F_1 = \frac{1}{k} \times \sum_{i=1}^k \frac{2 \times P_i \times R_i}{P_i + R_i} \quad (12)$$

where  $k$  is the number of true clusters. The higher the value of  $F_1$  is, the better the classification performance is.

##### 3. NMI

NMI [53] is commonly used in community detection to measure the similarity of two communities, as well as in overlapping clustering. Given two clustering results  $X$  and  $Y$ , NMI is calculated as:

$$NMI = \frac{I(X : Y)}{\max(H(X), H(Y))} \quad (13)$$

where  $H(X)$  ( $H(Y)$ ) is the entropy of the random variable  $X$  ( $Y$ ) related to the partition,  $I(X : Y)$  is the mutual information and derived based on  $H(X)$  and  $H(Y)$ . Detailed formula can be seen in [53]. The higher the value of  $NMI$  is, the more similar  $X$  and  $Y$  is.



**Table 2**

Different clustering performance on 10 datasets.

| Dataset     |       | CE3-kmeans                  | 3W-DBSCAN     | CE3-kmeans            | 3W-DBSCAN     | DScale-DBSCAN |
|-------------|-------|-----------------------------|---------------|-----------------------|---------------|---------------|
|             |       | Lower bound $\underline{C}$ |               | Upper bound $\bar{C}$ |               |               |
| 3L          | ACC   | 0.5014                      | <b>0.6982</b> | 0.5675                | <b>0.9411</b> | 0.8875        |
|             | NMI   | 0.1275                      | <b>0.4691</b> | 0.1541                | <b>0.7163</b> | 0.5582        |
|             | $F_1$ | 0.4939                      | <b>0.7628</b> | 0.5681                | <b>0.9189</b> | 0.8964        |
| 4C          | ACC   | 0.5536                      | <b>0.7592</b> | 0.6267                | <b>0.9232</b> | 0.8784        |
|             | NMI   | 0.2743                      | <b>0.5642</b> | 0.3097                | <b>0.7050</b> | 0.5668        |
|             | $F_1$ | 0.6148                      | <b>0.7896</b> | 0.6574                | <b>0.9072</b> | 0.8840        |
| S1          | ACC   | 0.7515                      | <b>0.9450</b> | 0.9102                | <b>0.9958</b> | 0.9762        |
|             | NMI   | 0.6686                      | <b>0.9019</b> | 0.8707                | <b>0.9800</b> | 0.9130        |
|             | $F_1$ | 0.8108                      | <b>0.9711</b> | 0.8845                | <b>0.9950</b> | 0.9863        |
| IRIS        | ACC   | <b>0.7932</b>               | 0.7200        | 0.8543                | <b>0.9800</b> | 0.8867        |
|             | NMI   | 0.5240                      | <b>0.5578</b> | 0.6247                | <b>0.8124</b> | 0.5928        |
|             | $F_1$ | 0.8154                      | <b>0.8273</b> | 0.8476                | <b>0.9555</b> | 0.9289        |
| Glass       | ACC   | 0.4311                      | <b>0.4673</b> | <b>0.4954</b>         | 0.4860        | 0.4673        |
|             | NMI   | 0.1123                      | <b>0.1918</b> | 0.1035                | 0.1506        | <b>0.1722</b> |
|             | $F_1$ | 0.3766                      | <b>0.3783</b> | 0.3786                | <b>0.3885</b> | 0.3675        |
| Seeds       | ACC   | <b>0.8451</b>               | 0.4524        | 0.8928                | <b>0.9429</b> | 0.7524        |
|             | NMI   | <b>0.5462</b>               | 0.2974        | 0.5818                | <b>0.7009</b> | 0.3393        |
|             | $F_1$ | <b>0.8657</b>               | 0.6157        | 0.8809                | <b>0.9252</b> | 0.8426        |
| Pathbased   | ACC   | 0.7268                      | <b>0.8767</b> | 0.7685                | <b>0.9933</b> | 0.9767        |
|             | NMI   | 0.4617                      | <b>0.7548</b> | 0.4474                | <b>0.8861</b> | 0.8830        |
|             | $F_1$ | 0.7327                      | <b>0.9324</b> | 0.7320                | 0.9775        | <b>0.9816</b> |
| Aggregation | ACC   | 0.7856                      | <b>0.9365</b> | 0.7913                | <b>1.0000</b> | <b>1.0000</b> |
|             | NMI   | 0.6677                      | <b>0.9324</b> | 0.6688                | <b>0.9724</b> | 0.9114        |
|             | $F_1$ | 0.7389                      | <b>0.9788</b> | 0.7372                | 0.9916        | <b>1.0000</b> |
| Compound    | ACC   | 0.6024                      | <b>0.9098</b> | 0.6403                | <b>0.9424</b> | 0.9373        |
|             | NMI   | 0.6067                      | <b>0.8300</b> | 0.5559                | <b>0.8456</b> | 0.8395        |
|             | $F_1$ | 0.5781                      | <b>0.5835</b> | 0.8964                | 0.9163        | <b>0.9174</b> |
| Flame       | ACC   | 0.8400                      | <b>0.8542</b> | 0.8484                | <b>1.0000</b> | 0.9708        |
|             | NMI   | 0.4505                      | <b>0.6674</b> | 0.4505                | <b>0.9488</b> | 0.7863        |
|             | $F_1$ | 0.8385                      | <b>0.9093</b> | 0.8432                | <b>0.9927</b> | 0.9804        |

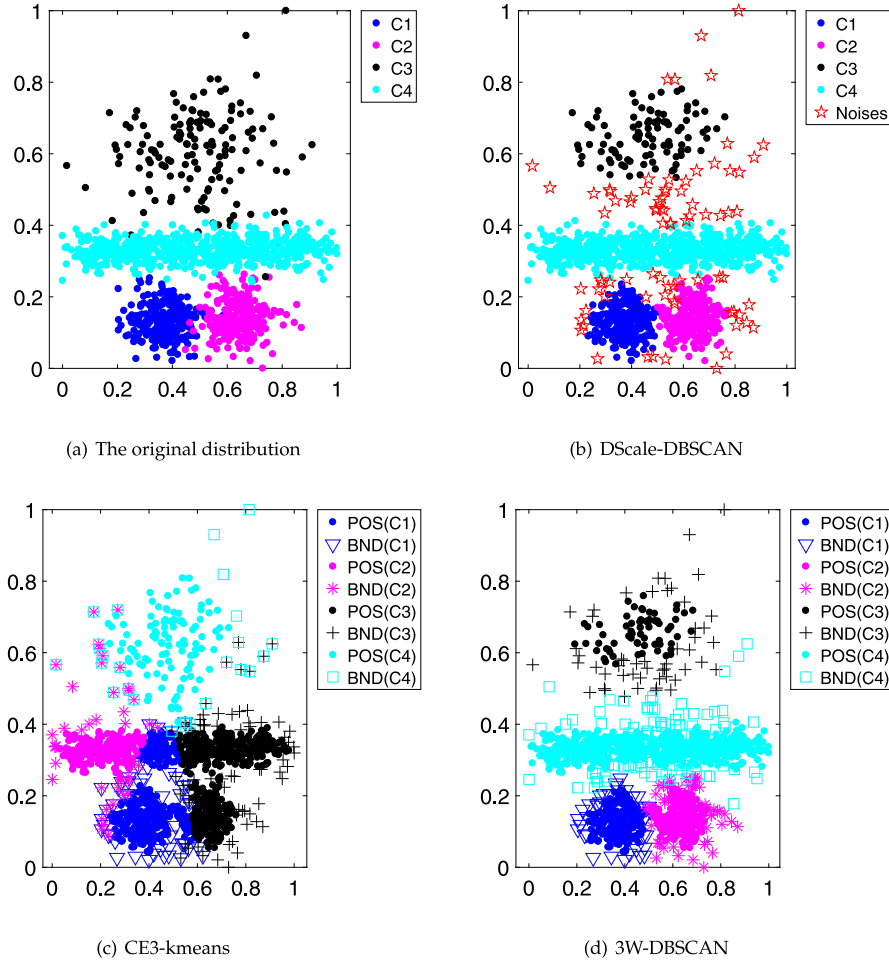
### 3.3. Performance of 3W-DBSCAN

We compare the performance of 3W-DBSCAN with the newly proposed CE3-kmeans [17] and DScale-DBSCAN [49] clustering algorithms, in which DScale-DBSCAN uses DScale for data processing and then implements DBSCAN to cluster. We do not choose other clustering algorithms because CE3-kmeans has compared the clustering effect with HCM [54], SC [55], FCM [56], RFCM [57], ECM [58], RECM [59] and DScale-DBSCAN has compared with DBSCAN [40], OPTICS [44] and DP [46]. We compute three indices values by using lower and upper bounds of three-way clustering respectively, i.e.,  $C = [\underline{C}, \bar{C}]$ , positive regions of all clusters comprise the lower bound, while the union of positive and boundary regions forms the upper bound. Considering that DScale-DBSCAN is a two-way clustering method, its clustering result is regarded as the performance on upper bound in three-way clustering. Besides, since the result of CE3-kmeans is different in every experiment, its overall performance is measured by an average value in 100 times runs. The clustering evaluation is provided in Table 2. The best performances in lower bound and upper bound for each dataset are highlighted in bold.

As shown in Table 2, both 3W-DBSCAN and CE3-kmeans give a lower Acc on lower bound and a higher Acc on upper bound for almost all datasets, as well as NMI and  $F_1$ . The reason can be attributed to the fact that each cluster is divided into a lower bound and an upper bound, these two sets are obtained by shrinking and expanding some elements respectively based on two-way clustering. It is reasonable that the clustering performance in upper bound is better than that in lower bound. Additionally, most Acc and NMI values obtained by DScale-DBSCAN are between the values on lower bound and upper bound by 3W-DBSCAN. This is because that based on the clustering result by DBSCAN, 3W-DBSCAN shrinks each cluster by picking out core objects, which constitutes the lower bound. Then 3W-DBSCAN expands each cluster by further processing with border and noise elements, which constitutes the upper bound. This process will improve the clustering performance.

In terms of algorithm, from Table 2, we can find that 3W-DBSCAN gives better performance than CE3-kmeans and DScale-DBSCAN in most cases, especially in synthetic and shape datasets, which demonstrates the superiority of 3W-DBSCAN. Take 4C and Compound as examples to illustrate the clustering performance in Figs. 5–6, note that the





**Fig. 5.** Four different schematic diagram of 4C.

color of class is different because class labels in four graphs are not the same. As we can see, owing to the intrinsic limitation of k-means, 3W-DBSCAN can detect clusters with arbitrary shape more accurately than CE3-kmeans. In addition, 3W-DBSCAN also has a relatively good performance on Compound, which has a distribution of different densities, it shows the ability of 3W-DBSCAN to handle with clusters with varied densities. Compared with DScale-DBSCAN, 3W-DBSCAN is capable to handle border and noise points, hence it is superior to DScale-DBSCAN to some extents.

However, we can also find that in the dataset IRIS, CE3-kmeans gives a higher  $Acc$  in lower bound but a lower  $F_1$  and  $NMI$  in upper bound than 3W-DBSCAN, a similar conclusion can be drawn in the datasets Seeds. In order to explain this phenomenon, we visualize the IRIS dataset and further analyze it. Note that IRIS has four features and Fig. 7 is the visualization of IRIS in the third and fourth dimensions. From Fig. 7(a), we can see that most points in truth cluster  $C_2$  are in the green slide elliptical area. In Fig. 7(c), CE3-kmeans can find almost all points of  $C_2$  and assign them into positive regions, while the positive region detected by 3W-DBSCAN (the dotted green elliptical area in Fig. 7(d)) only includes partial points of  $C_2$ , hence it is reasonable that CE3-kmeans gives a higher  $Acc$  than 3W-DBSCAN in lower bound. Furthermore, in the dotted green elliptical area of Fig. 7(c), CE3-kmeans assigns several objects into wrong clusters during the clustering process of  $C_2$  and  $C_3$ , leading to a lower  $NMI$  and  $F_1$  than 3W-DBSCAN. For the clustering performance of IRIS in upper bound, 3W-DBSCAN takes more objects into consideration (the solid green elliptical area in Fig. 7(d)), so the results of three evaluation indices are better than CE3-kmeans. As to the result of DScale-DBSCAN, it is better than CE3-kmeans but worse than 3W-DBSCAN in upper bound region. From Fig. 7(b), we can see that since the result of DScale-DBSCAN has many noisy points, so it cannot get a better performance than 3W-DBSCAN. It performs better than CE3-kmeans because CE3-kmeans assigns several objects into wrong clusters.

As seen in the Pathbased, Aggregation and Compound datasets in Table 2, 3W-DBSCAN gives a higher  $Acc$  and  $NMI$  than DScale-DBSCAN in upper bound, while DScale-DBSCAN gives a higher  $F_1$  than 3W-DBSCAN. The reason is that there

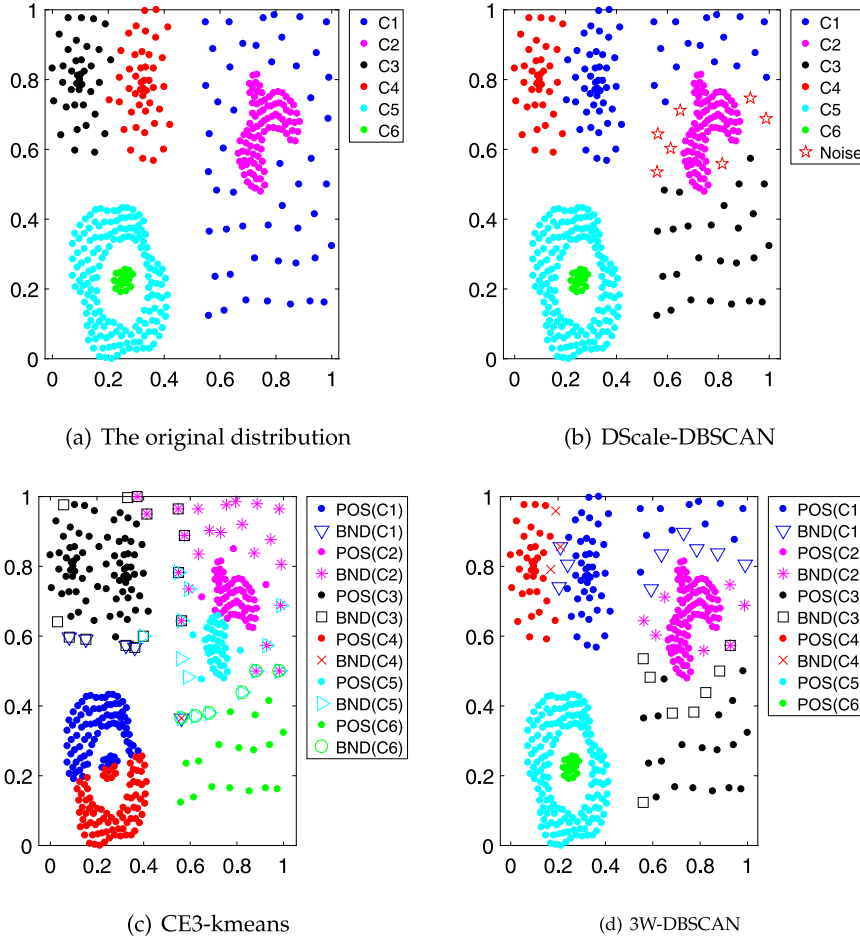


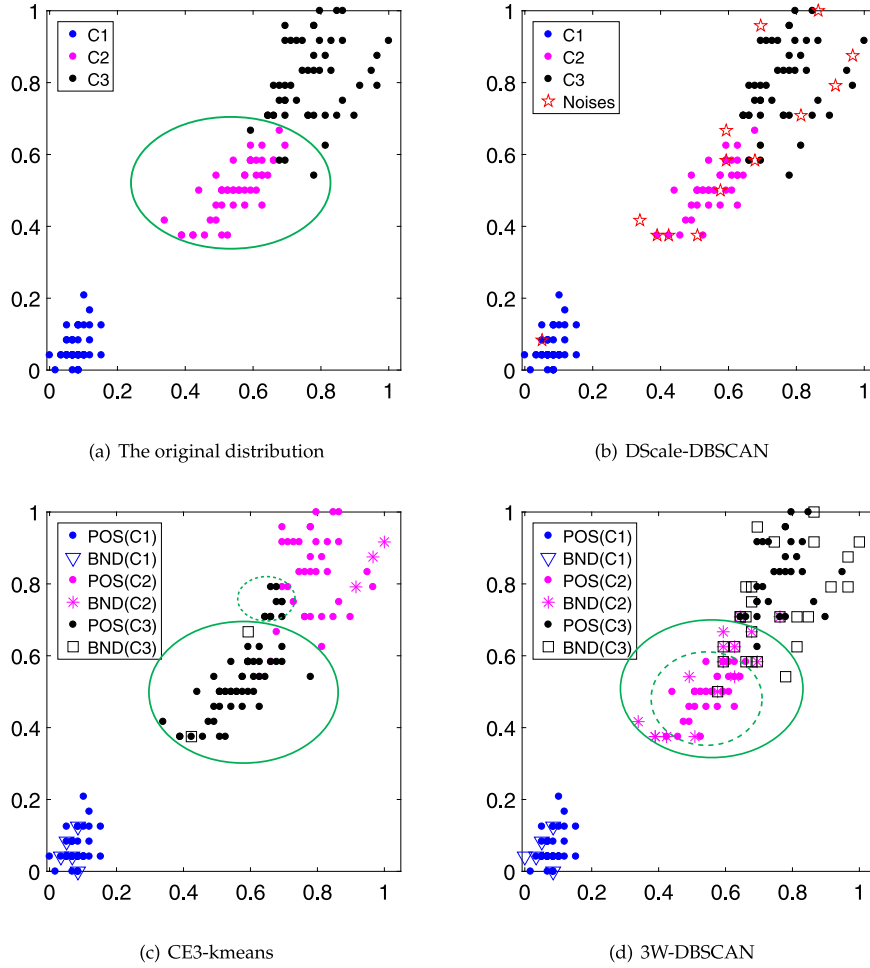
Fig. 6. Four different schematic diagram of Compound.

exist overlapping regions in the result of 3W-DBSCAN. Take Aggeration as an example. Fig. 8(a) is the original distribution of Aggeration dataset. It has two attributes and can be divided into seven clusters. As shown the elliptical area depicted by green solid lines in Fig. 8(d), we can find that the connected areas between  $C_4$  and  $C_5$ ,  $C_2$  and  $C_3$  are relatively compact, hence objects in these areas are regarded as overlapping elements belonging to two clusters simultaneously. While from the point of the true dataset (Fig. 8(a)), they just belong to one cluster strictly and thus they are misclassified by 3W-DBSCAN. Although the precision rate of 3W-DBSCAN is improved, the recall rate is declined, which finally leads to a slight decrease of  $F_1$ . Actually, as shown in Fig. 9 (the local enlarged drawing of overlapping clusters in Fig. 8(d)), we believe that to assign these points (the areas marked with blue rectangle lines in Fig. 9) into overlapping clusters is reasonable according to the visualization, although this assignment causes  $F_1$  to decrease. The phenomenon is also consistent with human's cognitive thinking.

Besides, we have evaluated the parameter  $\eta$  for improved DBSCAN, Fig. 10 shows the best  $F_1$  on upper bound for five datasets when  $\eta$  ranges from 0.1 to 0.5. It shows that the scaling effect is different from different datasets. However, generally, when  $\eta$  is set at around [0.15, 0.3], it is more likely to get the best clustering performance.

#### 4. Conclusion

In many applications of clustering, there exist objects whose relationship with clusters is ambiguous because they may belong to one or more clusters. To address the problem this paper provides a Three-Way clustering method based on an improved DBSCAN algorithm (3W-DBSCAN). Instead of a single set to express a cluster in two-way clustering, each cluster is described by a pair of nested sets called lower bound and upper bound respectively. It is consistent with human cognitive thinking and can get better results. By conducting experiments on 10 datasets, we compare the



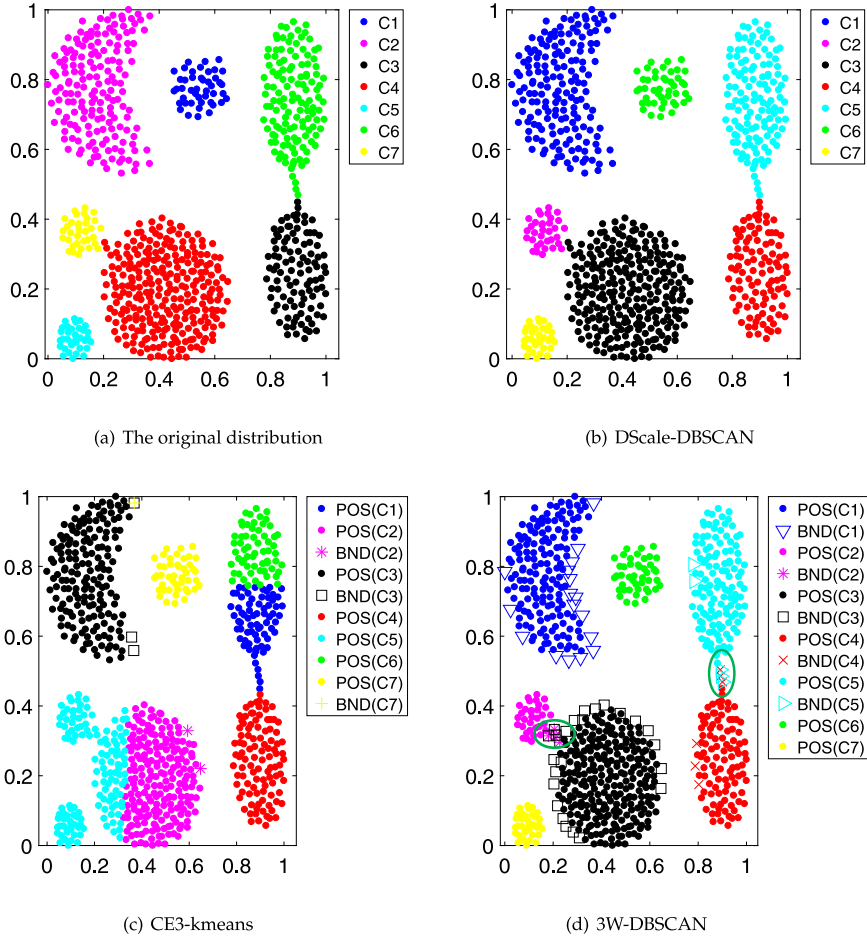
**Fig. 7.** Four different schematic diagram of IRIS. Note in the lower left quarter of (c) and (d), the seemingly overlapping sample between the positive and boundary regions is actually two samples which have the same feature values.

clustering performance on Accuracy ( $Acc$ ), F-measure ( $F_1$ ) and  $NMI$  evaluation indices with two new approaches CE3-kmeans and DScale-DBSCAN. The experimental results show that 3W-DBSCAN gives better performance in most cases, besides, using the visualization of datasets, the reason why 3W-DBSCAN fails to give good clustering results in some cases is also discussed.

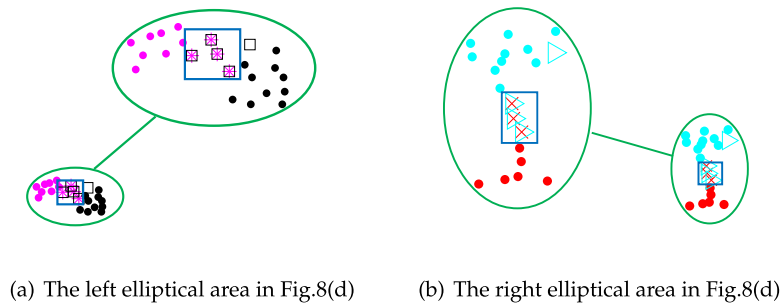
The following two topics will be studied as future research. First, the present study extracts the results of a hard clustering algorithm into three-way clusters, so the proposed method will depend on the performance of hard clustering algorithm to some extents, how to construct three-way clusters from scratch is a promising study. Second, the parameter  $\eta$  in 3W-DBSCAN has a great impact on clustering results, it is necessary to study an operability method to obtain it. What is more, the relationship between  $\eta$  and  $\epsilon$  will be an interesting topic.

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**Fig. 8.** Four different schematic diagram of Aggeration.



**Fig. 9.** The overlap region of Aggeration.

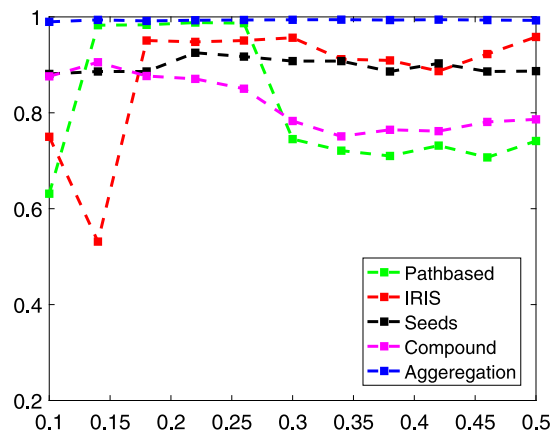


Fig. 10.  $F_1$  on four datasets with different  $\eta$  values.

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